Assignment 0. Warmup exercises Marks 20

Posted on 06.08.2026 @ 2:30 pm and due on 06.08.2026 @ 5:40 pm

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Attempt any four

1. Calculate the sum of first 20 even numbers and factorial of 8. (using `do-while` or `for` or any other loop). **[5]**

2. Calculate the sum of $N = 15$ terms of a GP and HP series for common difference 1.5 and common ratio 0.5 starting from $t_0 = 1.25$. Use of analytical formulae is not allowed. **[5]**

3. Consider the following matrices,

$$\mathbf{A} = \begin{pmatrix} 2 & -3 & 1.4 \\ 2.5 & 1 & -2 \\ -0.8 & 0 & 3.1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 & -1 & 1 \\ 1.5 & 0.5 & -2 \\ 3 & 0 & -2 \end{pmatrix}$$
$$\mathbf{C} = \begin{pmatrix} -2 \\ 0.5 \\ 1.5 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

   Find $\mathbf{AB}$, $\mathbf{D} \cdot \mathbf{C}$ and $\mathbf{BC}$. **[5]**

4. In 3-dimension Cartesian coordinate system with axes $(x, y, z) \equiv (0, 1, 2)$, the permutation matrix that exchanges $\mu \leftrightarrow \nu$ (where μ, $\nu = x, y, z$ or $0, 1, 2$) is given by

$$\left(A_{\mu\nu}\right)_{ij} = \delta_{ij} - \delta_{i\mu}\delta_{j\mu} - \delta_{i\nu}\delta_{j\nu} + \delta_{i\mu}\delta_{j\nu} + \delta_{i\nu}\delta_{j\mu}$$

   where $i, j = 0, 1, 2$. Determine A_{xy} *i.e.* A_{01}, A_{zy}, A_{xz}. **5**

5. Define *MyComplex* class / structure and calculate the sum, difference, product and modulus of $(1.3 - 2.2j)$ and $(-0.8 + 1.7j)$. **[5]**